

# Hermitian flavor sources and exact one-loop mass-phase cancellation

A complete canonical fermion action, finite hierarchical matching and an exact covariant determinant

PerfectPower research draft / 6 October 2026 (UTC)

Termwise one-loop protection with arbitrary real scalar mixing. Explicit source-center inputs, locally positive scalar EFT and a complete mass-only affine candidate calculation.

## The unified canonical result

A real Hermitian scalar nonet supplies a useful interaction constraint: every scalar variation of the heavy-to-light mixing block remains Hermitian. With a scalar universal heavy mass and Higgs Yukawa, the complete renormalizable fermion action then has a positive tree determinant and an exactly vanishing one-loop determinant phase. The cancellation holds for arbitrary real neutral-scalar mixing, including Higgs/flavor portals. The same canonical action realizes six hierarchical light masses and four finite-scale charged-current targets. No field-dependent quark metric is used in that benchmark.

The mass and current construction uses explicit fitted real source-center coefficients. It establishes compatibility of hierarchy, weak CP and one-loop mass-phase cancellation in one action; it does not derive the golden CKM relation. The scientific receipt retains those coefficients, all 128 branches of a separate mass-only affine search, and twelve fixed-mass deformations. The latter identify exactly where predictive freedom survives the one-loop theorem.

## A low-degree invariant basis with an exact determinant

$$\det \mathcal{W} = \chi^3, \quad \chi = \text{Im Tr}(R^2 T^2 R T)$$

Let  $R = A A^\dagger / r_{\text{scale}}$  and  $T = L L^\dagger / t_{\text{scale}}$  in the up-family triplet space, with fixed positive normalization constants. Both transform by conjugation. The following nine Hermitian matrix words have real coefficients and obey the elementary generalized CP transformation without an inserted imaginary unit:  $I$ ,  $R$ ,  $R$ -squared,  $T$ ,  $RT+TR$ ,  $R$ -squared  $T+T$   $R$ -squared,  $T$ -squared,  $RT$ -squared+ $T$ -squared  $R$ , and  $RT$ -squared  $R$ . Their elementary scalar degrees are respectively 0,2,4,2,4,6,4,6,8.

Use coordinates consisting of three real diagonal entries and the real and imaginary parts of entries 12,13,23. At a nondegenerate  $R$  eigenbasis define  $\Delta = (r_2-r_1)(r_3-r_1)(r_3-r_2)$  and  $\tau = \text{Im}(T_{12} T_{23} T_{31})$ . The determinant of the nine-word coordinate matrix is  $\Delta$ -cubed times  $\tau$ -cubed. Equivalently, if  $\chi = \text{Im Tr}(R\text{-squared } T\text{-squared } RT)$ , then  $\chi = \Delta \tau$  and the coordinate determinant is  $\chi$ -cubed. With orthonormal Hilbert-Schmidt off-diagonal coordinates the determinant is  $8 \chi$ -cubed. Nonzero  $\chi$  therefore provides a complete real Hermitian source-center basis through scalar degree eight.

The symbolic certificate factorizes the diagonal Vandermonde determinant  $\Delta$ , two off-diagonal coefficient determinants  $-\Delta$  and  $+\Delta$ , and three real pair determinants  $-\tau, +\tau, -\tau$ . The row-reordering sign yields the stated result. A generic symbolic Hermitian  $T$  also verifies  $\chi = \Delta \tau$ . An independent exact integer fixture has  $\Delta = 48$ ,  $\tau = 11$  and determinant 147197952. Generic multiprecision fixtures verify covariance, CP and the determinant identity.

The first eight displayed words have rank eight on the exact fixture; the last word lifts it to nine. This is a constructive result within the displayed matrix-word class, not a completeness claim for every finite-group covariant at lower degree. General covariant rank saturation and its implications for flavor predictivity are established results of Grinstein, Lu, Merlo and Quilez, arXiv:2312.13349. The present contribution is the explicit determinant certificate, low-degree real word basis and its application to this canonical interaction. A positive quark metric can likewise be completed as  $(I + \sum c_a Q_a)\text{-squared} + \eta I$  for  $\eta$  positive, with elementary scalar degree at most sixteen; such higher kinetic terms are allowed, rather than eliminated by the new construction.

## Complete renormalizable fermion interactions

Keep the certified product-family scalar representations and their existing phase charges. Ordinary quarks and one additional vectorlike weak-singlet quark triplet per sector transform as the up-family 3 of 3.A6. Introduce two independent shaping symmetries  $Z2_{up}$  and  $Z2_{down}$ . The vectorlike triplet of sector  $f$  has both chiralities odd under  $Z2_f$ , with ordinary quarks even. Introduce a real scalar nonet  $H_f$  in  $1+8$  of the up-family group, and a real scalar singlet  $\eta_f$ ; both are odd under  $Z2_f$  and neutral under the other shaping factor. The nonet is intrinsically Hermitian, with CP acting as  $H_f \rightarrow X H_f^* X^\dagger$ . It is not a complex nonet with a Hermitian vacuum imposed afterwards.

The five independent real fermion coefficients per sector multiply the ordinary Higgs Yukawa  $y_f$ , the vectorlike mass  $m_f$ ,  $\eta_f$  times the heavy-left/ordinary-right bilinear,  $\text{Tr}(H_f)/3$  times that bilinear, and the traceless  $H_f$  times that bilinear. Exact triplet endomorphism multiplicity one and adjoint-triplet contraction multiplicity one justify the singlet and adjoint contractions. The receipt enumerates the ten two-sector terms and their shaping charges. These are all nonzero in the benchmark:  $y_{up} = 1.2$ ,  $y_{down} = 0.57$ ,  $m_{up} = m_{down} = 1$ ,  $g = 0.1$ ,  $c_s = 0.7$  and  $c_a = 0.8$ .

The shaping charges forbid a Higgs coupling to a heavy right-handed triplet, a constant heavy-left/ordinary-right mixing, the opposite-sector nonet or  $\eta$  insertion, and a nonet or  $\eta$  insertion in the vectorlike mass. Existing elementary L,A,B,S representations supply no further single-insertion contraction on these endpoints. Bare canonical kinetic terms are universal within each irreducible triplet; an ordinary/heavy cross kinetic term is forbidden by the shaping charge. Positive constant normalization factors can be absorbed into the five real coefficients.

Write  $C_f = g \eta_f I + c_s \text{Tr}(H_f) I/3 + c_a (H_f - \text{Tr}(H_f) I/3)$ . It is Hermitian for every scalar configuration and every real choice of the five coefficients. In ordinary/heavy ordering the six-state Dirac matrix has blocks  $D_f = [[v y_f I, 0], [C_f, m_f I]]$ . Its determinant is  $(v y_f m_f)^3$ -cubed, positive on the chosen positive real branch and independent of the source vacuum. This is the Nelson-Barr triangular determinant mechanism, not a new principle of tree strong-CP protection.

## Exact finite spectra and physical currents

$$h_i^2 = \frac{(a^2 - \ell_i^2)(m^2 - \ell_i^2)}{\ell_i^2}, \quad M_i = \frac{am}{\ell_i}$$

Diagonalize  $C_f = U_f \text{diag}(h_i) U_f^\dagger$ . Applying  $U_f$  on both chiralities leaves three real two-state blocks  $[[a, 0], [h_i, m]]$ , where  $a = v y_f$ . For any chosen positive light mass  $\ell_i$  smaller than both  $a$  and  $m$ , the exact inverse relation is  $h_i^2 = (a^2 - \ell_i^2)(m^2 - \ell_i^2)/\ell_i^2$ . Either sign of  $h_i$  is allowed. The heavy partner mass is  $a m / \ell_i$ , without a small-flavon or small-vev expansion.

The ordinary-doublet weight obeys  $q_i^2 = a^2 - \ell_i^2$  ( $m^2 - \ell_i^2$ )/( $a^2 m^2 - \ell_i^4$ ). Consequently the finite light charged current is  $\text{diag}(q_{up}) U_{up}^\dagger U_{down} \text{diag}(q_{down})$ . It need not be unitary. The calculation retains all six fermions per sector, the neutral currents and positive light-current row and column deficits. Real coefficient changes in the five-term fermion action preserve the source eigenframes while altering masses and doublet weights; this statement excludes mass-ordering crossings and changes of the scalar source centers.

Use  $v = 0.03$  and illustrative up Yukawas  $(10^{-5}, 0.003, 0.9)$ , down Yukawas  $(2 \times 10^{-5}, 0.0004, 0.02)$ . These are test inputs, not measured running quark masses. Four target current quantities are obtained from PDG 2026 central unitary parameters  $\sin(\theta_{12})=0.22517$ ,  $\sin(\theta_{13})=0.003763$ ,  $\sin(\theta_{23})=0.04189$  and  $\delta=1.154$ :  $|V_{us}| = 0.2251684057717295$ ,  $|V_{cb}| = 0.04188970341421036$ ,  $|V_{ub}| = 0.003763$  and  $J = 3.159371628927255 \times 10^{-5}$ . Their use is explicit input matching.

The source eigenframe angles compensate the exact doublet weights, so those four quantities match in the finite current, rather than only in a unitary decoupling approximation. The largest relative current residual is  $2.67 \times 10^{-15}$ . The up and down mass residuals are below  $1.95 \times 10^{-13}$  and  $6.89 \times 10^{-15}$ . The CP-conjugate source produces  $-J$  with zero recorded double-precision partner residual. All source targets are realized by the nine real polynomial coefficients at 90-digit working precision. The largest coefficient magnitudes are approximately  $8.08 \times 10^{10}$  and  $2.00 \times 10^{10}$ . These large declared inputs preclude treating the construction as a demonstrated natural or perturbative scalar UV completion.

## An exact one-loop phase theorem

$$G_{IK}G_{KI} \in \mathbb{R} \implies \text{Im Tr}(D^{-1}\delta D) = 0$$

In the source eigenbasis the left and right fermion rotations are real two-state matrices. Every physical neutral real-scalar mode has a Hermitian source variation  $A$ , plus a real flavor-diagonal Higgs vertex. A mass-basis source vertex connecting states  $(i, \sigma)$  and  $(j, \tau)$  therefore equals a real two-state factor times  $A_{ij}$ . For  $i$  different from  $j$ , its chirality-flipping paired product  $G_{IK}G_{KI}$  is a real factor times  $|A_{ij}|^2$ . Within a single flavor,  $A_{ii}$  and every Higgs contribution are real. Every paired product is thus real separately, including interference between the Higgs and nonet portions of an arbitrarily mixed mode.

The one-loop diagonal chirality-changing self-energy is a sum of these paired products times real masses and real loop weights. Its imaginary part vanishes term by term. Hence  $\text{Im Tr}(D^{-1}\delta D) = 0$  exactly for every real neutral-scalar mixing matrix and positive scalar spectrum, without a small spurion or a cancellation between sectors. The same argument cancels the UV phase coefficient. Generic tests with 91 independently mixed modes give only floating-point errors of order  $10^{-15}$ ; these errors are checks of the analytic identity, not a predicted residual  $\theta$ .

Neutral gauge phase factors use Hermitian left currents and real diagonal right currents. The full charged and neutral Goldstone Ward products reduce to real squared-current magnitudes. Their reality is checked with all six states per sector. Charged gauge loops contribute Hermitian wavefunction factors. Scalar tadpole shifts preserve Hermitian  $C$  and the identically real tree determinant. These observations close the first-order one-loop mass-phase cancellation for the specified complete renormalizable fermion action and canonical bare metrics.

This theorem concerns the first-order one-loop determinant phase. Higher fermion or field-dependent kinetic vertices and two-loop corrections are outside its assumptions. It is not an all-loop strong-CP solution. One-loop-vanishing Nelson–Barr extensions already exist in the literature; see Alves and Nishi, arXiv:2506.03257 / JHEP 09 (2025) 162. No priority claim is made for that general possibility. The concrete result here is the real-nonnet interaction, termwise proof with arbitrary scalar mixing, and exact finite hierarchy/current construction in the same canonical action.

## A locally positive 91-scalar branch

Start from the portal-shifted 71-scalar Higgs/flavor stationary branch at  $v = 0.03$ . The four Higgs norm portals are  $10^{-4}$ ,  $2 \times 10^{-4}$ ,  $3 \times 10^{-4}$  and  $4 \times 10^{-4}$ . Its verified minimum scalar mass squared is  $9.40405681 \times 10^{-7}$  and maximum scaled gradient is  $9.64 \times 10^{-12}$ . For each sector form  $F_f = \sum c_{fa} Q_a(R,T)$ , using the explicit fitted real coefficients, and add  $\text{Tr}(H_f - \eta_f F_f)^2 / 2 + (\eta_f^2 - 1)^2$  to the base scalar potential.

At  $\eta_f = 1$  and  $H_f = F_f$  all new squares vanish. The old stationary point and its effective base potential are retained. This interaction respects the shaping charges and CP. Its maximum elementary scalar degree is eighteen; it is a specified scalar EFT, not a renormalizable scalar UV completion. The two nonets add eighteen real components and the two eta fields add two, giving 91 physical neutral real scalars.

The full quadratic form is  $\delta x^T A_0 \delta x + \text{norm}(\delta H - J \delta x)^2$ , where  $x$  consists of the old 71 coordinates and the two eta coordinates,  $A_0$  is the direct sum of the verified old Hessian and  $8I$ , and  $J$  is the source-center Jacobian. This congruence proves positivity for every finite  $J$ . A conservative conditional eigenvalue lower bound is  $\min(\lambda_{\min}(A_0), 1) / (1 + \text{norm}(J))^2$ , evaluated as  $4.70 \times 10^{-31}$ . The tiny bound reflects the large fitted coefficients; it is not a numerically resolved 91-state eigenvalue. Local positivity is conditional on the verified base Hessian and does not establish global or quantum vacuum selection. The one-loop theorem itself holds for arbitrary scalar mixing and does not depend on this particular squared-lock potential.

## Fixed-mass freedom inside the protected action

Rotate the down source by each of six independent anti-Hermitian flavor generators with either sign of a  $10^{-4}$  displacement, and recalibrate its nine real center coefficients. All twelve branches keep the old base vacuum stationary, preserve local positivity by the same congruence, retain the five real fermion coefficients, and retain all six finite light masses within  $6.40 \times 10^{-14}$  relative error. The tree and one-loop determinant-phase identities remain valid on every branch.

The four-observable response for  $(|V_{us}|, |V_{cb}|, |V_{ub}|, J)$ , after fractional response and column normalization, has singular values approximately  $(2.0097, 1.1586, 0.5964, 0.5131)$ . Thus all four current directions remain independent at fixed masses even while one-loop strong CP is protected. This is a concrete separation of strong-CP protection from CKM predictivity. The source-center coefficients, and the allowed higher kinetic basis, must be constrained by additional justified interactions before an observable golden relation can follow.

## A complete mass-only affine candidate calculation

$$q^3 f(t)^2 - d^2 g(t)^3 = 0, \quad t = a_2/a_1$$

A stricter candidate retains only  $C_f = a_0 I + a_1 R + a_2 T$ . At the fixed source vacuum and chosen Higgs/vectorlike coefficients, the six mass inputs determine a finite set of source centers without any CKM input. Let  $R_0, T_0$  denote traceless  $R, T$ , let  $c_i$  be the centered signed target eigenvalues, and set  $q = \sum c_i^2$  and  $d = \text{product } c_i$ . Define  $g(t) = \text{Tr}(R_0^2) + 2t \text{Tr}(R_0 T_0) + t^2 \text{Tr}(T_0^2)$ , and  $f(t) = \det R_0 + t \text{Tr}(R_0^2 T_0) + t^2 \text{Tr}(R_0 T_0^2) + t^3 \det T_0$ . With  $t = a_2/a_1$  the exact spectral constraint is  $q^3 f(t)^2 - d^2 g(t)^3 = 0$ .

This sextic is computed over rational numbers from explicitly declared decimal input fixtures. Exact Sturm counts and rational isolating intervals of width at most  $10^{-35}$  certify every real root. The reconstruction chooses  $a_1 = \text{sign}(d/f) \sqrt{q/g}$ ,  $a_2 = t a_1$ , and adjusts  $a_0$  to the mean eigenvalue. Pure-T branches at infinity are separately checked and absent on the fixture. Four inequivalent signed eigenvalue patterns are included; an overall source sign leaves masses and current magnitudes unchanged. This equivalence is observable, rather than an assertion that all signs lie on one vacuum group orbit.

There are eight up branches and sixteen down branches, yielding 128 branch pairs. Their full finite currents, unitary decoupling charts, CP quartets and root intervals are retained. Eighty-digit source reconstruction checks accompany every branch. None reproduces the nominated  $|V_{us}|$  near 0.225 on this mass/source fixture; branches with  $|V_{us}|$  between 0.1 and 0.4 instead lie near 0.1444–0.1611. This excludes this precisely specified affine candidate, not other source potentials or the full framework. The restriction to three source words is not enforced against the other allowed center or kinetic operators. Its completed exhaustive calculation supplies a controlled baseline for testing a future interaction constraint.

## Reproduction and scientific scope

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_hermitian.py`, then `PYTHONPATH=python python python/tests/test_valentiner_hermitian.py`. The release has eighteen focused checks and fifty-six related quantum, canonical, adjoint and kinetic checks. Fresh replay reproduces the scientific receipt byte for byte. Exact polynomial certificates, approximate scalar-vacuum inputs and fitted coefficient inputs are kept distinct. The standalone paper and cumulative monograph include this chapter in full.

References: Grinstein, Lu, Merlo and Quilez, arXiv:2312.13349; Alves and Nishi, arXiv:2506.03257, DOI 10.1007/JHEP09(2025)162; PDG, CKM Quark-Mixing Matrix (2026), equation 12.28. Primary URLs are retained in the scientific receipt. The calculations establish the stated conditional results; publication priority and a predictive physical UV completion require independent assessment.